

## Machine Tool Technology Week-7

### 1. Explain Thread cutting Mechanism

#### a) Half Nut Mechanism

#### b) Tumbler Gear Mechanism

### 2. Prepare the Process plan as per the given drawing for Thread cutting

### 3. Estimate the Production cost per Piece as per the given drawing for Thread cutting (Consider all Direct and Indirect costs)

### Thread Cutting Mechanism

#### Use of Lead Screw with half nut:

- The lead screw is used for thread cutting.
- It is driven from the headstock through the feed gearbox and moves the carriage in a longitudinal direction against the work-piece.
- From the lead screw, the motion is transmitted to the carriage by engaging the half nut with the lead screw.
- Acme threads are used in lathe lead screws.

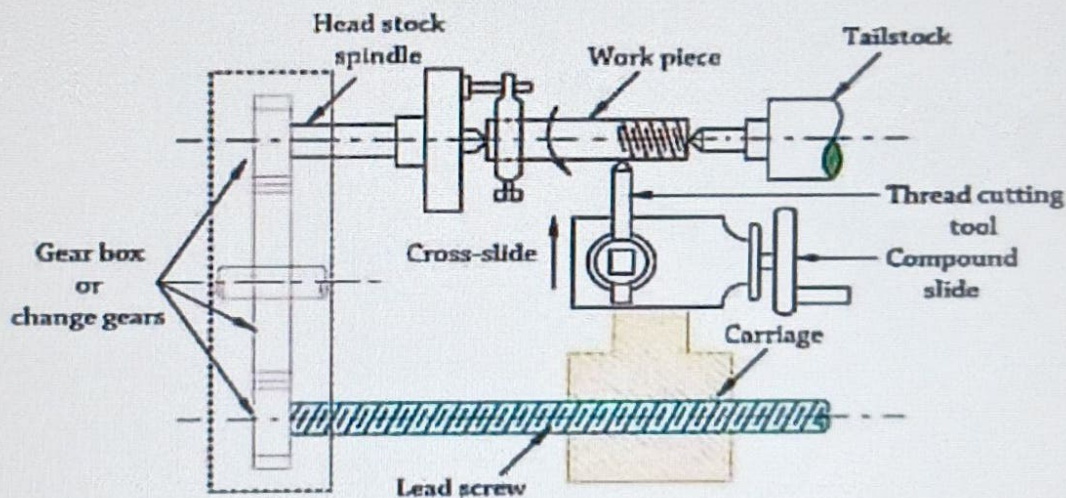
This form of thread enables the easy engagement of the half nut.

- A half nut is used to engage the lead screw to the carriage so that the carriage moves longitudinally in a direction parallel to the axis of rotation when the lead screw rotates through the spindle motor.

The half nut or split nut is used for thread cutting in a lathe.

It engages or disengages the carriage with the lead screw so that the rotation of the leadscrew is used to traverse the tail along the workpiece to cut screw threads.

The direction in which the carriage moves depends upon the position of the feed reverse lever on the headstock.



**Lathe set-up for thread cutting operation**

### **Tumbler Gear Mechanism**

In a lathe, the tumbler gear mechanism is a simple gear train, typically consisting of three gears, used to change the direction of rotation of the lead screw and feed shaft, enabling the carriage to move in either direction for threading or general turning.

#### **Purpose:**

The primary function of the tumbler gear mechanism is to reverse the direction of rotation of the feed rod, which in turn controls the movement of the carriage (tool holder) along the lathe bed.

#### **Components:**

**Fixed Stud Gear:** This gear is mounted on a fixed stud and is driven by the spindle gear.

**Tumbler Gears:** These gears are mounted on a bracket that can be moved to engage either of the tumbler gears with the fixed stud gear.

**Driven Gear:** This gear is connected to the feed rod and is driven by the tumbler gears.

#### **Operation:**

The spindle drives the fixed stud gear, which in turn drives the tumbler gears.

By moving the tumbler bracket, you can engage either tumbler gear with the fixed stud gear, thereby changing the direction of rotation of the driven gear and, consequently, the direction of the carriage movement.

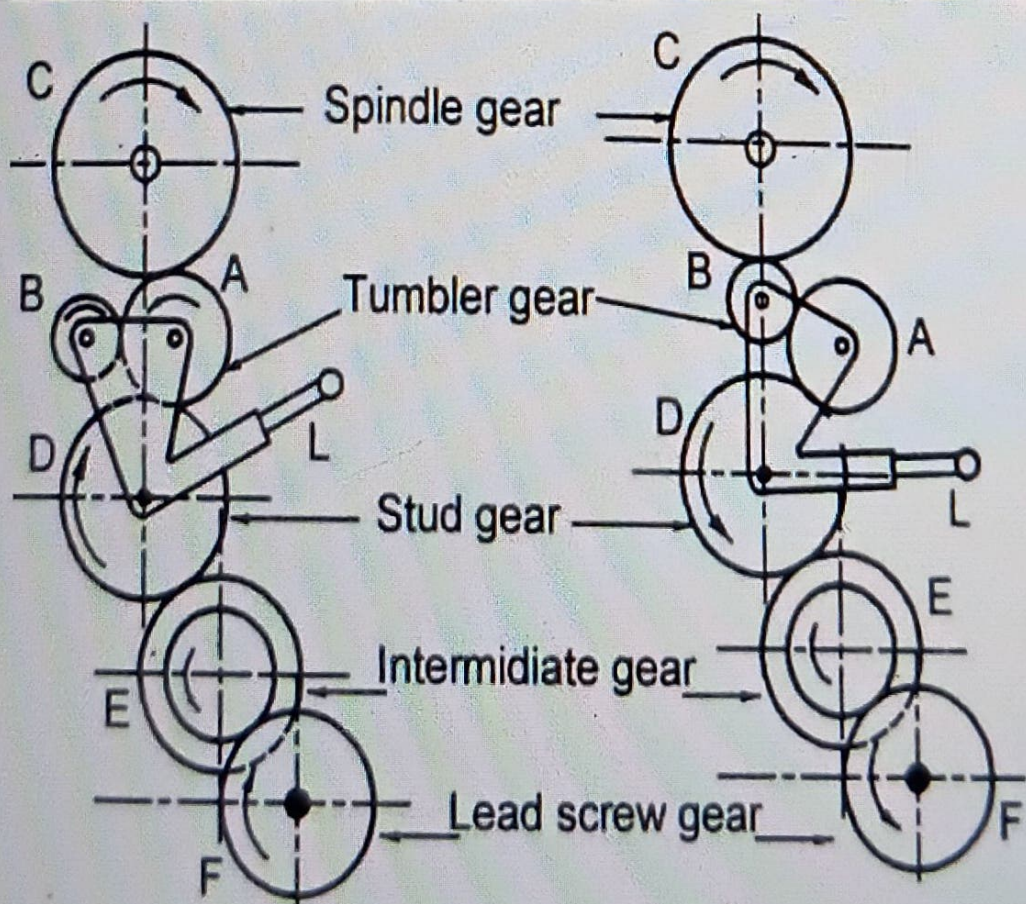
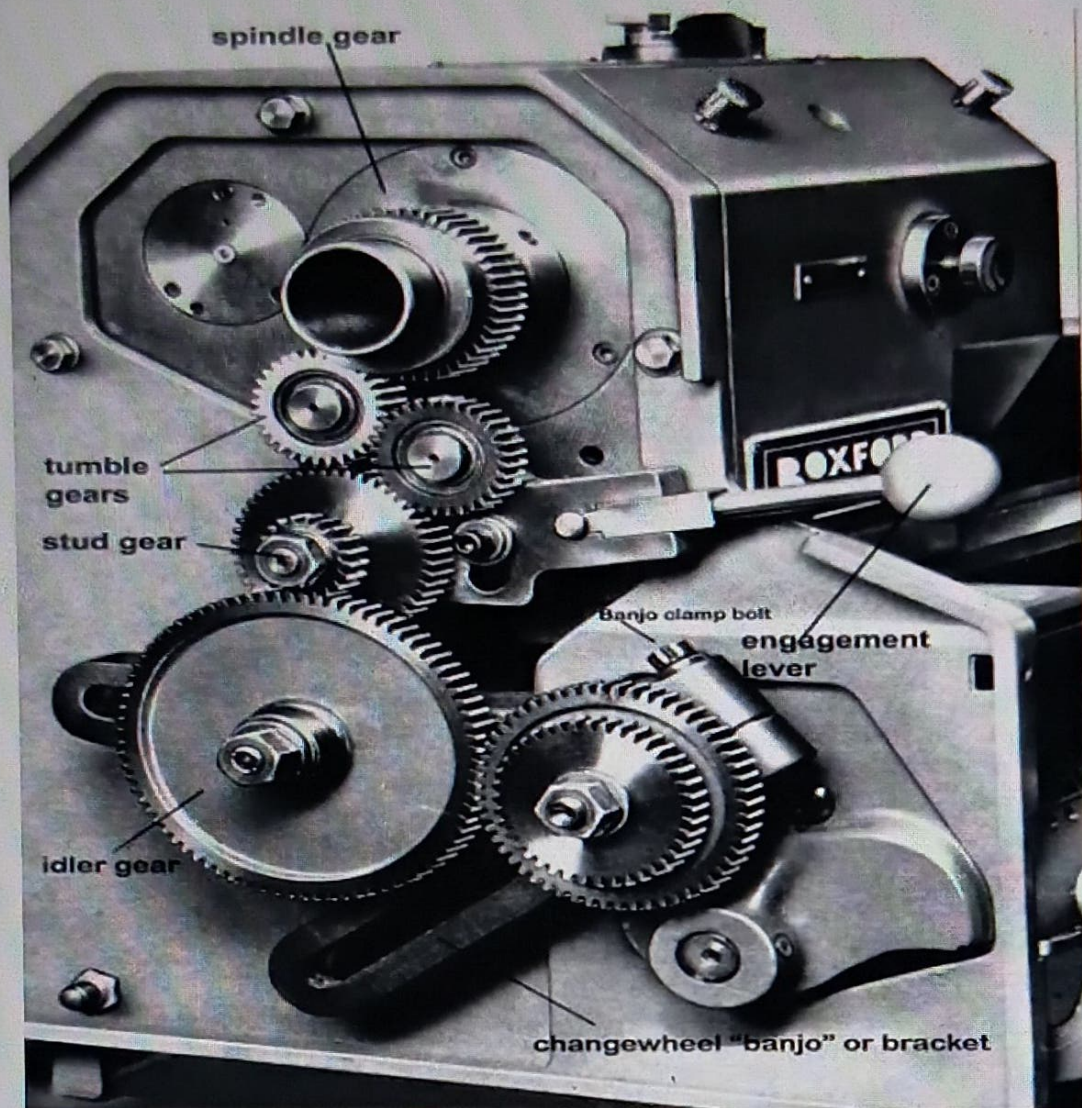
#### **Location:**

The tumbler gear mechanism is typically located between the spindle drive and the feed gearbox.

#### **Applications:**

**Threading:** Tumbler gears are essential for threading operations, allowing the carriage to move in the correct direction for cutting threads.

**General Turning:** They can also be used for general turning operations, allowing the carriage to move in either direction as needed.



To perform thread cutting on a given work piece of mild steel as per dimensions

### TOOLS REQUIRED:

- 1) Marking Block
- 2) Tool key
- 3) Chuck key
- 4) Single point cutting tool
- 5) Thread cutting tool

### MEASURING INSTRUMENTS REQUIRED:

- 1) Steel rule
- 2) Out – side calipers
- 3) Vernier calipers
- 4) Pitch Gauge

**MATERIAL REQUIRED:** A mild steel rod of length 100 x  $\phi$  25mm piece

### OPERATIONS:

- 1) Fixing the job in the chuck
- 2) Fixing single point cutting tool in the tool post
- 3) Facing
- 4) Plain turning
- 5) Step turning
- 6) Thread cutting

### PROCEDURE:

1. With the help of marking block and chuck key, the job is fixed in the chuck so that the axis of job coincides with the axis of the lathe.
2. The cutting tool is fixed in the tool post with spanner.
3. Facing is done by giving cross feed with cross slide at one end.
4. Marking is done along the work piece where steps are to be made.
5. By giving longitudinal feed and depth of cut steps are made according to dimensions.
6. For doing thread cutting speed of work should be as low as possible. It is done by changing pulley belts and by engaging back gear.
7. In the gear train we have to arrange suitable gear wheels for setting the required pitch using the formula given below.

8. By engaging the lead screw and half nut the thread cutting tool is made to move as per the pitch of the work piece on the job and required threading is done.
9. Using pitch gauge the obtained threading is inspected.
10. Chamfer is made.
11. The job is reversed and facing, plain turning and step turning is done as per the required dimensions using steel rule, outside calipers and vernier calipers.
12. Chamfer is made.

### CHANGE GEAR CALCULATIONS:

$$\frac{\text{Driver teeth}}{\text{Driven teeth}} = \frac{5p_n}{127}$$

Where P = Pitch of the work piece to be made in mm.  
And n = Pitch of the lead screw in mm

### PRECAUTIONS:

1. The job & the tool should be fixed tightly in the chuck & tool post respectively.
1. During operations work piece & tool should not be touched by hand.
2. Proper care should be taken so that the tool should not touch the chuck.
3. During thread cutting operation sufficient coolant oil should be poured to cool the work piece and tool.

**RESULT:** The job of required dimensions is obtained.

**Example 29.** A manufacturer desires to manufacture 100 bolts/day from mild steel bars of length 2 m. and dia. as 25 mm. Assume :

- (i) Feed for turning = 0.2 cm./rev.
- (ii) Cutting speed for turning = 20 m./min.
- (iii) Cutting speed for threading = 10 m./min.
- (iv) Depth of cut = maximum of 3 mm.
- (v) Pitch for threads 2.5 mm.
- (vi) No. of cuts required for threading = 6.

Calculate the machining time for 100 bolts, if the operator is paid at the rate of Rs. 1.25/hr. Estimate the labour charges, considering other allowances for fatigue etc. as 100%.

**Solution.**

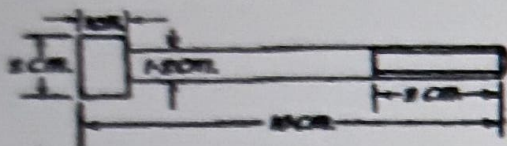


Fig. 6-15

### Step I. Turning

- (i) Reducing the dia. from 25 mm. to 20 mm. for a length of 10 cm.

Turning time

$$t_1 = \frac{\text{Length}}{\text{Feed/rev.} \times \text{r.p.m.}}$$

$$= \frac{10}{0.2 \times \frac{100 \times 20}{\pi \times 2.5}} = 0.2 \text{ min.}$$

- (ii) Reducing the dia. from 20 mm. to 12 mm. in two cuts a length of 9 cm.

Time required in first cut to reduce the dia. from 20 mm. to 16 mm.

$$t_2 = \frac{9}{0.2 \times \frac{100 \times 20}{\pi \times 2}} = 0.15 \text{ min.}$$

And time required in second cut

$$t_3 = \frac{9}{0.2 \times \frac{100 \times 20}{\pi \times 1.6}} = 0.12 \text{ min.}$$

∴ Total turning time

$$T_1 = t_1 + t_2 + t_3$$

$$= 0.20 + 0.15 + 0.12 = 0.47 \text{ min.}$$

### Step II. Threading.

Threading time for one cut

$$= \frac{\text{Length}}{\text{Pitch} \times \text{r.p.m.}}$$

$$= \frac{30}{2.5 \times \frac{100 \times 10}{\pi \times 1.2}} = 0.045 \text{ min.}$$

∴ Total threading time for 6 cuts

$$= 0.045 \times 6 = 0.27 \text{ min.}$$

∴ Total machining time

$$= 0.47 + 0.27 = 0.74 \text{ min.}$$

∴ Time required for 100 bolts

$$= 0.74 \times 100 = 74 \text{ min. Ans.}$$

∴ By considering allowances as 100%

Time taken by operator = 74 × 2

$$= 148 \text{ min.} = 2.46 \text{ hrs.}$$

∴ Labour cost = 2.46 × 1.25

$$= \text{Rs. 3.08. Ans.}$$